# SMART GAS CYLINDER MONITORING AND NOTIFICATION SYSTEM

## 1. INTRODUCTION

Liquefied Petroleum Gas (LPG) is a widely used household and industrial fuel, making gas level monitoring and safety a crucial concern. Often, users run out of gas unexpectedly, causing inconvenience and potential hazards due to leakage or overuse. Traditional methods of checking cylinder levels are manual and inaccurate.

To address this issue, this project proposes an IoT-based Smart Gas Cylinder Monitoring and Notification System that automatically measures the remaining gas weight and provides alerts when it reaches a predefined threshold. The system integrates a 40 kg Load Cell with an HX711 amplifier to accurately measure cylinder weight. The data is processed by an ESP32 microcontroller, which transmits real-time weight readings to the cloud via Wi-Fi. Users can monitor gas levels through a web or mobile dashboard and receive notifications when refilling is required.

This solution offers a low-cost, reliable, and scalable way to manage household gas usage efficiently while minimizing manual intervention.

## 2. AIM OF THE PROJECT

The main aim of this project is to design and implement an IoT-enabled gas cylinder monitoring system that continuously measures the gas weight, sends notifications when the gas level is low, and enables users to monitor it remotely.

## 3. OBJECTIVES

- To measure the gas cylinder's real-time weight using a load cell and HX711 module.
- To process sensor data through an ESP32 microcontroller and transmit it to the cloud.
- To develop a user interface for real-time gas level monitoring.
- To send automatic alerts when the cylinder weight drops below a threshold.
- To provide a compact, safe, and cost-effective system suitable for household environments.

## 4. EXISTING SYSTEM

The existing methods for gas level monitoring are mostly manual or analog, such as tapping the cylinder or using mechanical pressure gauges.

**Disadvantages:**

- Manual checking is inaccurate and time-consuming.
- Pressure gauges only measure gas pressure, not the actual quantity.
- No real-time notifications or automation.
- Inefficient tracking of gas consumption trends.

## 5. PROPOSED SYSTEM

The proposed system introduces an IoT-based smart gas monitoring device capable of measuring the cylinder's weight accurately and sending data to a cloud server in real time.

The ESP32 microcontroller collects data from a load cell (via HX711 amplifier) and compares it with a predefined empty weight. When the cylinder's gas weight falls below a set threshold, the system sends an alert via Wi-Fi to the connected user dashboard or smartphone.

This design eliminates the need for manual checks, offering automation, accuracy, and convenience.

## 6. SYSTEM ARCHITECTURE

The architecture comprises two main layers — **Device Layer** and **Cloud Layer**.

### 6.1 Hardware Components

- **ESP32 DevKit V1:** Microcontroller with built-in Wi-Fi for processing and data transmission.
- **Load Cell (40 kg):** Measures the total weight of the cylinder.
- **HX711 Amplifier Module:** Converts the analog signal from the load cell into a digital signal for the ESP32.
- **Platform/Base Board:** A stable board to hold the cylinder and load cell assembly.
- **Jumper Wires:** For connecting components during prototyping.
- **Power Adapter (5V):** Powers the ESP32 board.

### 6.2 Software Components

- **Arduino IDE:** Used to code and upload programs to the ESP32.
- **Flask + MongoDB (Cloud):** For real-time data storage and communication.
- **Dashboard Interface:** Displays live cylinder status and notifications.
- **Docker :** For containerized backend deployment.

## 7. WORKING PRINCIPLE

1. **Initialization:** The ESP32 initializes Wi-Fi and connects to the HX711 module.
2. **Data Acquisition:** The load cell continuously measures the total cylinder weight.
3. **Data Processing:** The ESP32 subtracts the empty cylinder + board weight to determine gas content.
4. **Threshold Check:** When the measured weight falls below the set threshold, a "Low Gas" alert is triggered.
5. **Notification:** Data is sent to the cloud and displayed on the dashboard or app.
6. **Monitoring:** Users can view real-time gas levels and refill alerts through the dashboard.

## 8. CONNECTION DETAILS

| Component | Pin Type | ESP32 Pin | Voltage |
|---|---|---|---|
| HX711 VCC | Power | 3.3V | 3.3V |
| HX711 GND | Ground | GND | 0V |
| HX711 DT | Data | D21 | 3.3V |
| HX711 SCK | Clock | D22 | 3.3V |

## 9. CLOUD & CONTAINER HOSTING

The backend server can be hosted on **Microsoft Azure** or any cloud service using Docker containers for scalability.

- **Azure VM:** Ubuntu-based VM with open ports (22, 5000, 3000).

- **Docker Setup:** Containerized services for Flask API, MongoDB, and dashboard.
- **Deployment:** Automated via docker-compose for ease of scaling.
- **Data Management:** Sensor data stored in MongoDB and visualized via dashboard.

## 10. RESULTS

The implemented system successfully:

- Measures real-time gas cylinder weight.
- Sends low-gas alerts when the gas level drops below threshold.
- Displays live weight readings on the cloud dashboard.
- Enables remote monitoring from any device connected to the internet.

## 11. ADVANTAGES

- Accurate real-time gas level monitoring.
- Automated notifications reduce manual effort.
- Easy to install and low cost.
- Can be expanded for multiple cylinders.
- Reduces unexpected gas exhaustion and improves safety.

## 12. APPLICATIONS

- **Domestic Use:** Monitor LPG usage in households.
- **Restaurants:** Continuous monitoring for multiple cylinders.
- **Industrial Kitchens:** Prevent downtime during high-demand operations.
- **Distribution Centers:** Track gas levels for refilling logistics.

## 13. FUTURE ENHANCEMENTS

- **Integration with Gas Booking API:** Automatically book refills when gas is low.
- **Mobile App Notifications (SMS/Push):** Real-time alerts to smartphones.

- **Voice Assistant Integration:** Works with Alexa or Google Assistant.
- **Leak Detection:** Adding a gas sensor for safety monitoring.
- **Solar Power Support:** For standalone operation.

## 14. CONCLUSION

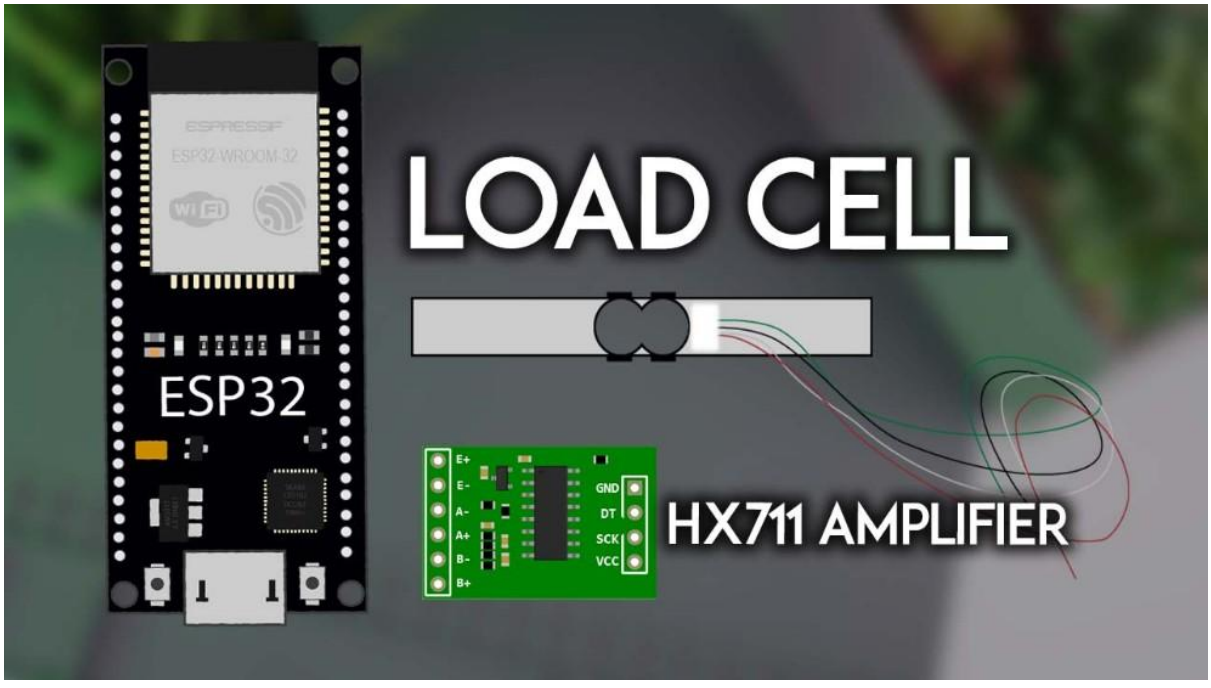
The **Smart Gas Cylinder Monitoring and Notification System** provides an intelligent, automated solution to track LPG levels accurately. It ensures timely alerts, improves user convenience, and enhances safety. The IoT integration makes the system scalable and accessible via the cloud, laying the foundation for future smart home automation applications.

## 15. PROJECT DESIGN

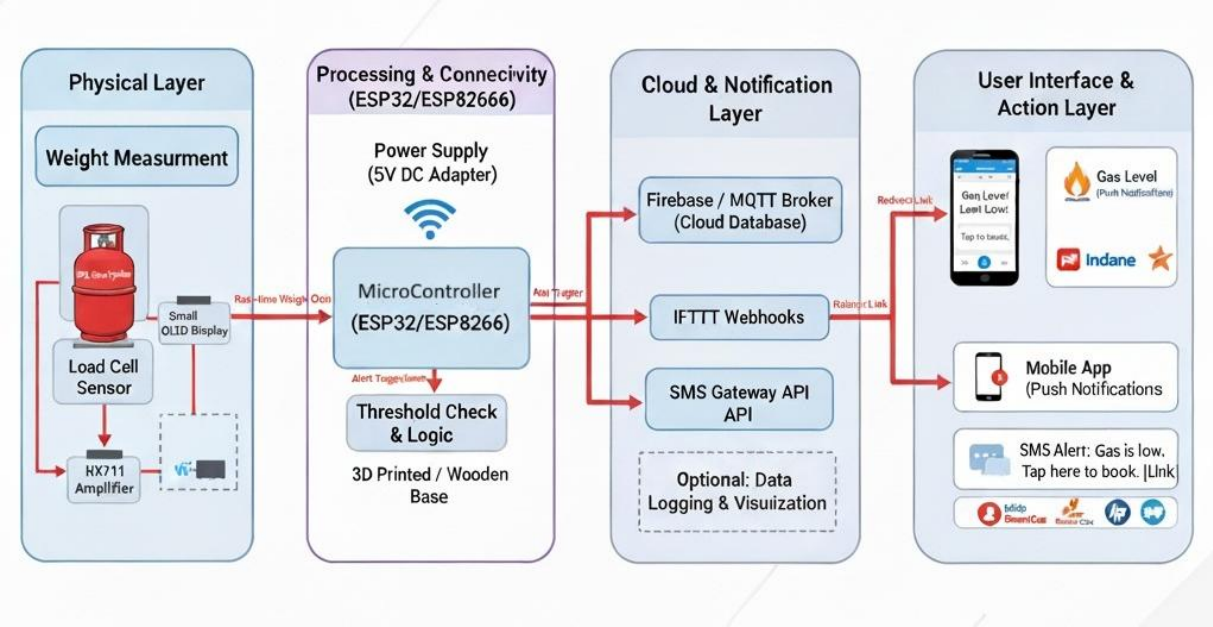
**Data Flow:**

1. Load Cell → HX711 → ESP32: Measures and digitizes gas weight.
2. ESP32 → Cloud: Transmits weight data via Wi-Fi.
3. Cloud → Dashboard/App: Displays current weight and alerts.

## 16. HARDWARE COMPONENTS



## 17. BLOCK DIAGRAM




## 18. FINAL PRODUCT AND PRESENTATION





## 19. DASHBOARD

